

Information Projection and Selective Search: Implement Now or Draw Again*

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Abstract

We study how interpersonal information projection bias influences a researcher’s decision to implement a current idea or to continue drawing new ones. Consistent misbeliefs, such as overconfidence or underconfidence, distort beliefs in a fixed manner. In contrast, information projection creates state-dependent beliefs: before an idea arises, the researcher is optimistic about its uniqueness, and once the idea is in hand, the researcher becomes pessimistic. This fluctuation makes implementation less attractive and makes repeated searches more appealing. Our analysis yields two key findings. First, compared to any consistent-misbelief researcher whose fixed belief falls between the optimistic and pessimistic assessments of the information projection researcher, the information projection researcher is more selective: implementing less, drawing more ideas, and quitting less frequently. Second, at sufficiently high performance thresholds, researchers who end up passing the threshold are disproportionately information projection researchers. Consequently, information projection systematically biases both individual search dynamics and the composition of such researchers.

JEL Classification: D83, D91

Keywords: Information projection bias, Over/underconfidence, Selectiveness, Implement or draw ideas

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1 Introduction

Researchers often face a fundamental dilemma: upon encountering a promising idea, should they implement it now or discard it to draw a new one? The decision is complicated not only by uncertainty about the idea’s true value but also by concerns about uniqueness, specifically, whether others may have already had the same idea. This concern is amplified in competitive environments and by the “file drawer problem,” where null or negative results are less visible than published successes (Rosenthal, 1979). Thus, second-order beliefs about others’ knowledge are central to the decision to implement or continue searching.

This paper studies how a specific bias, *interpersonal information projection*, shapes this decision compared to standard distortions such as overconfidence or underconfidence. Information projection bias suggests that a person believes others know what she or he knows and do not know what she or he does not know (e.g., Madarász, 2012). In our context, information projection produces a distinctive, state-dependent distortion. Before an idea occurs, the researcher is optimistic about its uniqueness, believing that others are unlikely to have it. However, once she has the idea, her perspective shifts, and she becomes pessimistic, thinking that others are likely to have it.¹ Thus, information projection stands in sharp contrast to biases such as overconfidence or underconfidence, which are modeled as fixed, time-invariant distortions of perceived uniqueness.

We formalize this in a stationary search environment. Ideas arrive sequentially, each with an informative signal, and implementation is a one-shot decision: once used, the idea yields a single outcome and is exhausted. Instead of implementing the existing idea, researchers can draw a new idea and move to the subsequent implementation decision. Researchers may also choose to quit, which yields a normalized payoff of zero. We compare two types of researchers with different beliefs. A *consistent-misbelief (CM)* researcher maintains a fixed subjective probability that another researcher, whom we call a *precursor*, has already had the idea. By contrast, an *information projection (IP)* researcher applies a lower perceived precedence before an idea arises

¹We use the female pronoun for a researcher and the male pronoun for a precursor referenced later.

and a higher perceived precedence once the idea is in hand, but fails to anticipate this shift.²

Our analysis yields two main results. First, we establish a behavioral distinction between the two belief types. Compared to any CM researcher whose fixed belief lies between the IP researcher’s optimistic ex-ante belief and pessimistic ex-post belief, the IP researcher is *more selective*: less likely to implement the current idea and more likely to draw again. The mechanism is straightforward: ex-post pessimism decreases the value of the current idea, while ex-ante optimism inflates the value of future draws. The wedge between “now” and “later” makes implementation comparatively unattractive and quitting less attractive. This result implies that information projection tilts researchers to search for new ideas persistently.

Second, we present a selection result. For sufficiently high-performance thresholds, such as those for prizes, hiring, or promotion, researchers who end up passing the threshold are disproportionately IP researchers. We refer to such researchers as accomplished researchers. Intuitively, higher implementation cutoffs imply fewer implementations but at higher signal realizations. Under standard regularity conditions (the monotone likelihood ratio property and a standard tail regularity condition), this raises the eventual success probability and tilts the accomplished pool toward IP researchers. Thus, interpersonal information projection not only shapes individual behavior but also affects the composition of accomplished researchers.

Related Literature Our paper contributes to three strands of work. First, within the literature on projection bias, we focus on *interpersonal* information projection rather than *intrapersonal* projection (e.g., Loewenstein, O’Donoghue, and Rabin, 2003; Kaufmann, 2022). Following Madarász (2012) and subsequent applications (e.g., Danz, 2020; Madarász, 2021; Madarász, Danz, and Wang, 2023; Bushong and Gagnon-Bartsch, 2024),³ the source of projection in our model is another agent’s *knowledge* of the idea occurring ex ante, instead of tastes (e.g., Gagnon-Bartsch, 2016; Gagnon-Bartsch, Pagnozzi, and Rosato, 2021; Gagnon-Bartsch and Rosato, 2022).

²We focus on this type of *naïve* IP in the main text. For completeness, we discuss the case of a *sophisticated* IP type that anticipates its own belief shift in Section 3.2 and analyze it formally in Appendix B.

³These studies apply information projection to models different from ours: Danz (2020) develops a market-entry model; Madarász (2021) considers an alternating-offer bargaining model; Madarász, Danz, and Wang (2023) employ a static principal–agent model; and Bushong and Gagnon-Bartsch (2024) consider a team-production model.

This distinction matters for behavior: information projection makes beliefs *state-dependent*: optimistic about idea uniqueness before the idea is drawn and pessimistic after it is in hand. By contrast, standard overconfidence or underconfidence imposes a single, time-invariant distortion. Technically, this state dependence appears as a wedge between the belief used to value the current idea and the belief used to value continuation (future ideas), which is the key driver of our predictions on the draw-implement margin.

Second, relative to research on overconfidence in search, our environment builds in uncertainty about the *value of stopping* (implementing) and delivers opposite comparative statics. Dubra (2004) shows that overconfidence lengthens search in an ordinal model in which the stop value is known; in our model, because the stop value is uncertain and filtered through signals, overconfidence or underconfidence shifts both current payoffs and continuation values. In contrast, information projection generates a larger wedge: it lowers the perceived value of implementing the current idea (ex-post pessimism) while raising the perceived value of continuing to search (ex-ante optimism). This combination makes implementation less likely and quitting less attractive, a pattern we show does not arise for any consistent-misbelief type within the same belief interval.

Finally, we connect to experimentation and bandit models surveyed by Hörner and Skrzypacz (2017) and to selection in dynamic search. Unlike ambiguity-driven stopping in Auster, Che, and Mierendorff (2024), where ambiguity aversion induces short early searches and long later searches, our mechanism is preference-rational but subject to state-dependent belief bias: the same decision maker applies inconsistent beliefs before and after idea arrival in a non-Bayesian way. The consequence is a *higher* implementation threshold together with a *higher* continuation value, which in turn implies a selection result: among researchers who eventually clear a high performance threshold, information projectors are overrepresented. This selection channel is distinct from social-learning explanations of the file-drawer phenomenon (e.g., Rosenthal, 1979; Herrera and Hörner, 2013): while non-observation of outcomes is informative in both settings, our results are not driven by cascade dynamics. Instead, our mechanism is an IP-induced distortion of the draw-implement margin.

2 Model

Timing The researcher faces an infinite-horizon sequence of one-shot research ideas. Let $\delta \in (0, 1)$ be a discount factor. We index periods by $t = 0, 1, 2, \dots$. At $t = 0$, the researcher draws an idea. At the beginning of period t , if a draw was *initiated in* $t - 1$, its signal s arrives, and the information cost $k > 0$ is incurred.

Given the information arrives at the beginning of t , the researcher then chooses one of the following actions:

1. **Implement (one-shot).** Implementation uses period t . At the beginning of $t + 1$, the outcome v is realized and c is paid. Here, $c > 0$ is the cost of implementation. A new draw is *initiated in* $t + 1$; its signal arrives at the start of $t + 2$, and the information cost k is paid. Then, the implement/discard/quit decision is made.
2. **Discard and draw immediately.** The current idea is permanently removed and cannot be recalled. A new draw is *initiated in* t ; its signal arrives at the start of $t + 1$, and the information cost k is paid. Then, the implement/discard/quit decision is made.
3. **Quit.** The process concludes. The outside option is 0.

Each idea can be implemented at most once; implementation yields no flow payoffs beyond the single outcome. A discarded idea cannot be implemented later.⁴ Furthermore, whether the idea was previously implemented is known to the researchers.

Example. If the researcher implements at $t = 1$, she obtains the result and pays c at $t = 2$. She also initiates a new draw in $t = 2$. The next signal then arrives at the start of $t = 3$, k is paid, and the next decision is made. Alternatively, if she discards at $t = 1$, she initiates a new draw immediately in $t = 1$. The signal arrives at the start of $t = 2$, k is paid, and she makes her decision again.

We model an environment in which implementing an idea exhausts that idea. Thus, after implementation, if she continues, the researcher returns to the draw state for the next idea. One could

⁴This is without loss of generality in our stationary environment with no additional information arriving about previously seen ideas: any policy that discards an idea now and implements it later is weakly dominated by either implementing it now or never implementing it.

alternatively allow the researcher to implement the current idea and then decide in the following period whether to continue or exit. This alternative timing would modify the Bellman equation only by a constant term and would not affect our main comparison between CM and IP researchers. We adopt the current timing for expositional simplicity.

Signal and outcome After arriving at an idea, the researcher receives a signal $s \in \mathbb{R}$, which correlates with the actual value of that outcome when implementing the idea. The value of the outcome is $v \in \mathbb{R}$, realized if the idea remains available, that is, if it has not already been implemented by a precursor. v is uncertain and distributed by F , where F denotes the cumulative distribution function (CDF). The researcher gains payoff $u(v)$ from the research outcome v , where $u: \mathbb{R} \rightarrow \mathbb{R}_+$ is strictly increasing, continuously differentiable, and $\sup_{x \in \mathbb{R}} u(x) < \infty$.

The distribution of signal s depends on the true value v . $G(\cdot | v)$ denotes the CDF of the signal. We assume that all CDFs have full support on $\mathbb{R}$. Let f and g be the probability density functions of F and G , respectively. We assume that F and G are twice continuously differentiable for all variables. Additionally, we assume that, for each $v > v'$, $G(\cdot | v)$ dominates $G(\cdot | v')$ in the sense of the monotone likelihood-ratio property (MLRP).⁵

Precursors We assume that another researcher may have had the same idea that occurs to the researcher. We refer to such a previous researcher as a *precursor*. For each idea, with probability $p \in [0, 1]$, a precursor has already had it. In other words, with probability $1 - p$, the researcher is the first one to draw the idea. p is the objective probability and the researcher can have misbelief about this probability, as explained in the next paragraph, “Beliefs about precursors”.⁶ To simplify, we assume that the value of implementing the idea v is independent of whether a precursor had the idea, provided he had not implemented it.

Upon receiving his signal s , the precursor makes a one-shot implementation decision; if he implements, the realized outcome becomes public, which is known to all researchers. For simplicity,

⁵Equivalently, $\frac{g(s|v)}{g(s|v')}$ is increasing in s . In other words, $\frac{\partial^2 \ln(g(s|v))}{\partial s \partial v} > 0$.

⁶The objective probability p plays a direct role in Section 3.4, where accomplishment probabilities are evaluated under the true distribution.

precursors' decisions are independent of the researcher's dynamic problem, and we do not model their own precursors.⁷

Beliefs about precursors Ideas are indexed by $i = 1, 2, \dots$. For each idea i in the period when it occurs, the researcher forms a fresh belief about whether a precursor has already had the same idea, and beliefs are formed *idea by idea*. Depending on how these beliefs are formed, we consider two types of researchers.

Let ρ denote the researcher's subjective probability that a precursor has already had the idea. A *consistent-misbelief (CM) researcher* holds a constant subjective probability $\rho \in [0, 1]$ across every idea and period. When $\rho < p$ ($\rho > p$), the researcher is overconfident (underconfident) about the idea's uniqueness.

An *information-projection (IP) researcher* exhibits interpersonal information projection bias: the researcher believes that “others know what I know but do not know what I do not know.” Let $I_i \in \{0, 1\}$ indicate the information state, where $I_i = 0$ represents the state before having idea i , and $I_i = 1$ represents the state after having it. We denote the parameter of information projection bias by $\alpha \in (0, 1)$. The subjective probability that a precursor has already had the same idea is

$$p^{I_i} = (1 - \alpha)p + \alpha I_i, \quad \text{so} \quad p^{\text{pre}} := p^{I_i=0} = (1 - \alpha)p, \quad \text{and} \quad p^{\text{post}} := p^{I_i=1} = (1 - \alpha)p + \alpha,$$

hence $p^{\text{pre}} < p < p^{\text{post}}$. Thus, information projection generates ex-ante optimism about uniqueness and ex-post pessimism once an idea is at hand.

Unless otherwise noted, an IP researcher is *naïve*: she evaluates all future periods as if her belief would remain p^{pre} even after an idea arrives. Thus, the continuation value for future ideas is perceived to be stationary at p^{pre} . In contrast, the researcher uses the ex-post belief p^{post} for her current-period implementation decision. The agent does not anticipate this belief shift ex ante. We formalize this mapping in Section 3.2 and compare it with the corresponding mapping for a

⁷Modeling a one-shot precursor is sufficient for the analysis conditional on survival. Here, *survival* means that the idea has not been implemented by a precursor and therefore remains available to the researcher. This formulation keeps the exposition transparent.

sophisticated IP researcher who anticipates the future shift in Appendix B.

We deliberately abstract from Bayesian learning about p . In other words, we assume that the values of ρ and $(p^{\text{pre}}, p^{\text{post}})$ do not change throughout the game.

Finally, we assume that the precursor’s problem is one-shot. Although there might be several specifications of higher-order beliefs on others’ biases, we adopt this assumption for simplicity. Our objective is to focus on comparing the effects of first-order biases.⁸

Remark 1. Our treatment of information projection is reduced-form rather than a literal application of the signal-projection framework in Madarász (2012). In our setting, “having an idea” is not modeled as a primitive private signal. Instead, we use the notion of information projection to capture a state-dependent belief wedge about precedence: before an idea arrives, the researcher projects non-possession onto others, whereas after arrival she projects possession onto others. Additionally, our formulation implicitly assumes that the researcher does not project onto her future self her present lack of knowledge about ideas she has not yet drawn: she expects to recognize a valuable idea upon its arrival.

3 Analysis

3.1 Decision of Consistent-Misbelief Researcher

Each researcher’s decision depends on the precursor’s decision: The precursor solves a one-shot problem and implements if and only if $E[u(v) \mid s] \geq c$. Under the MLRP, $E[u(v) \mid s]$ is strictly increasing in s , hence there is a unique cutoff $\bar{s}$ such that $E[u(v) \mid \bar{s}] = c$. Conditional on the true value v , the probability that the precursor implements the idea is $1 - G(\bar{s} \mid v)$.

We next examine a CM researcher’s decision. We consider the expected utility of the implementation condition in the event that the implementation outcome of the drawn idea is not publicly known and that the researcher observes signal s . Let $\varphi_\rho(v, s)$ be the joint density of (v, s) conditional

⁸In the literature similar to ours, Madarász (2021) assumes that while one player exhibits information projection bias, the counterpart holds rational beliefs in analyzing an alternating offer bargaining model.

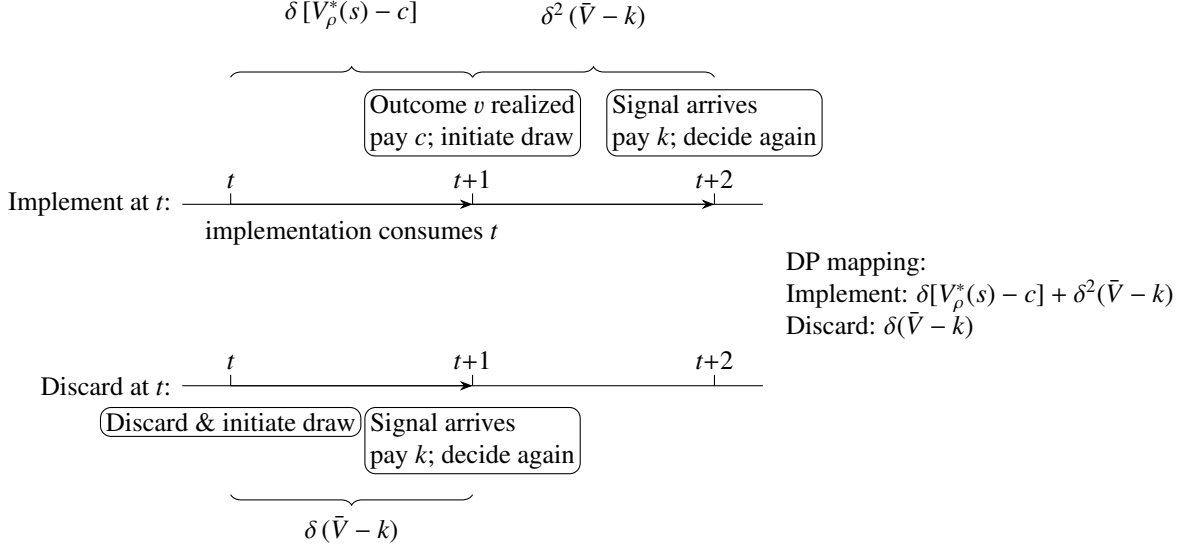


Figure 1: Timing. Implementing at t yields the outcome at $t+1$ (cost c); a new draw is initiated in $t+1$, so the next signal arrives at $t+2$ (cost k). Discarding at t initiates a draw immediately; its signal arrives at $t+1$ (cost k).

on survival of the idea with the true value v and the observed signal s :⁹

$$\varphi_\rho(v, s) := f(v)[(1 - \rho) + \rho G(\bar{s} | v)] g(s | v),$$

where $(1 - \rho) + \rho G(\bar{s} | v)$ is the researcher's perceived probability that the drawn idea has not been implemented by a precursor. The expected utility is

$$V_\rho^*(s) := \frac{\int u(v) \varphi_\rho(v, s) dv}{\int \varphi_\rho(v, s) dv}.$$

From the MLRP of G , we obtain the following lemma:¹⁰

Lemma 1. (i) $V_\rho^*(s)$ decreases in ρ ; (ii) $V_\rho^*(s)$ increases in s .

Dynamic program (DP) under a consistent belief ρ We first solve the dynamic problem conditional on staying active; the outside option to quit (value 0) is imposed afterward. Figure 1 reviews

⁹Note that whether the idea is implemented is not a state to be projected. As this probability depends on the utility maximization of the precursor, the belief ρ affects the probability through the precursor's behavior.

¹⁰Proofs of Lemmata 1, 3 and 4 and Proposition 1 are presented in Appendix A.

the timing of actions.

Let $\bar{V}$ denote the perceived continuation value upon entering next period's draw state, just before paying the information cost. Given a signal s , implementing now yields the expected payoff

$$\delta[V_\rho^*(s) - c + \delta(\bar{V} - k)],$$

whereas drawing a new idea yields

$$\delta(\bar{V} - k).$$

Hence the cutoff $s_\rho^*(\bar{V})$ is uniquely defined by

$$V_\rho^*(s_\rho^*(\bar{V})) = (1 - \delta)(\bar{V} - k) + c. \quad (1)$$

By Lemma 1, $V_\rho^*(s)$ is decreasing in ρ and increasing in s . Then, (1) yields the following lemma:

Lemma 2. $s_\rho^*(\bar{V})$ increases in ρ and $\bar{V}$.

Next, we define the continuation value $\bar{V}$ using a Bellman equation. When evaluating *future* ideas under belief ρ , the belief about future signal distribution is affected by ρ . To capture this, we define

$$\bar{g}_\rho(s) = \frac{\int f(v)[(1 - \rho) + \rho G(\bar{s} | v)] g(s | v) dv}{\int f(v)[(1 - \rho) + \rho G(\bar{s} | v)] dv}, \quad \bar{G}_\rho(s) = \int_{-\infty}^s \bar{g}_\rho(s') ds'.$$

Recall that $\bar{s}$ is the precursor's implementation cutoff defined by $E[u(v) | \bar{s}] = c$. $\bar{g}_\rho(s)$ is the probability density function of s conditional on the idea *not* having been implemented by a precursor, which implies the idea's survival, and $\bar{G}_\rho(s)$ is its CDF. Under the MLRP, we can show that $\bar{G}_\rho(s)$ is increasing in ρ .

Lemma 3. $\frac{\partial \bar{G}_\rho(s)}{\partial \rho} > 0$ for each $s \in \mathbb{R}$.

The intuition is as follows. As ρ increases, it becomes more likely that a precursor has already had the idea. Conditioning on survival then selects relatively lower signal s 's. Hence, the surviving

distribution of s shifts downward in the sense of first-order stochastic dominance.

Using the survival distribution, the continuation value solves the following Bellman equation:

$$\bar{V} = \delta \int_{s \geq s_\rho^*(\bar{V})} [V_\rho^*(s) - c + \delta(\bar{V} - k)] \bar{g}_\rho(s) ds + \delta(\bar{V} - k) \bar{G}_\rho(s_\rho^*(\bar{V})). \quad (2)$$

Intuitively, with probability $1 - \bar{G}_\rho(s_\rho^*)$, the researcher implements next period, obtaining $V_\rho^*(s) - c$ in that period and $\bar{V} - k$ in the following period, which yields $\delta[V_\rho^*(s) - c + \delta(\bar{V} - k)]$. With probability $\bar{G}_\rho(s_\rho^*)$, she redraws next period, obtaining $\bar{V} - k$, which yields $\delta(\bar{V} - k)$.¹¹ Since the value of drawing next period is $\delta(\bar{V} - k)$, the researcher quits if and only if $\bar{V} < k$.

Now we show the uniqueness of the solution to the Bellman equation, and perform comparative statics with respect to ρ .

Lemma 4. *Let $\bar{V}(\rho)$ solve (2). Then $\bar{V}(\rho)$ is uniquely determined. Furthermore, if $\bar{V}(\rho) > k$, then $\bar{V}(\rho)$ decreases in ρ .*

Intuitively, increasing ρ affects (2) in two ways. First, it lowers $V_\rho^*(s)$ from Lemma 1. Second, it shifts the *surviving* signal distribution: $\bar{G}_\rho$ increases pointwise with ρ from Lemma 3. In other words, survival places relatively more mass on low s when ρ is higher, which further reduces the right-hand side of (2). Both effects lower $\bar{V}$ as ρ rises.

Given $\bar{V}(\rho)$ and $s_\rho^*(\bar{V}(\rho))$, the period decision is characterized as follows. Because the researcher's belief is ρ in every period, she implements if $s > s_\rho^*(\bar{V}(\rho))$ and $\bar{V}(\rho) > k$; if $\bar{V}(\rho) > k$ but $s < s_\rho^*(\bar{V}(\rho))$, she discards the current idea and draws a new idea; and she quits if and only if $\bar{V}(\rho) < k$.

3.2 Decision of Information-Projection Researcher

We assume that an IP researcher is naïve: she evaluates all future ideas as if her belief about precursors remained at p^{pre} . The naïve IP researcher evaluates future search opportunities under

¹¹See also Figure 1, which illustrates the correspondence between (2) and the game timing.

the pre-arrival belief p^{pre} and does not anticipate that, once an idea arrives, she will switch to the post-arrival belief p^{post} . This naïveté about her own future belief shift is what separates the continuation value from the current implementation decision. Hence, the continuation value is the fixed point of (2) evaluated at belief p^{pre} . In contrast, the current implementation decision is evaluated under p^{post} . Thus, the researcher's plan is characterized by continuation value $\bar{V}(p^{\text{pre}})$ while the implementation decision is characterized by cutoff $s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}}))$. This clarifies why the Bellman equation involves a single belief parameter for continuation, yet the implementation threshold uses another for the current period.

Formally, the analysis is as follows. Let V^{IP} denote the unique fixed point under p^{pre} :

$$V^{\text{IP}} = \delta \int_{s \geq s_{p^{\text{pre}}}^*(V^{\text{IP}})} [V_{p^{\text{pre}}}^*(s) - c + \delta(V^{\text{IP}} - k)] \bar{g}_{p^{\text{pre}}}(s) ds + \delta(V^{\text{IP}} - k) \bar{G}_{p^{\text{pre}}}(s_{p^{\text{pre}}}^*(V^{\text{IP}})).$$

Equivalently, $V^{\text{IP}} = \bar{V}(p^{\text{pre}})$ is the continuation value upon entering the next period's draw state when future ideas are evaluated with p^{pre} .

In contrast, given a realized signal s , the current-period implementation payoff is evaluated with p^{post} as $V_{p^{\text{post}}}^*(s) - c$. Therefore, the IP researcher implements the current idea if and only if

$$V_{p^{\text{post}}}^*(s) \geq (1 - \delta)(V^{\text{IP}} - k) + c,$$

i.e., using the cutoff $s_{p^{\text{post}}}^*(V^{\text{IP}})$. As the naïve IP researcher does not anticipate this shift ex ante, the Bellman equation features a single belief (for continuation) while the implementation threshold uses another (for the current period).

In summary, by substituting $V^{\text{IP}} = \bar{V}(p^{\text{pre}})$, the researcher implements the current idea if and only if $s > s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}}))$ and $\bar{V}(p^{\text{pre}}) > k$; if $\bar{V}(p^{\text{pre}}) > k$ but $s < s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}}))$, she draws a new idea; and if $\bar{V}(p^{\text{pre}}) < k$, she quits.¹²

Remark 2 (Sophisticated IP). While we consider a naïve researcher, we can also consider a “so-

¹²We can show that when $p^{\text{pre}} = p^{\text{post}} = \rho$, the IP researcher's optimal choice collapses to the CM's one with $V^{\text{IP}} = \bar{V}(\rho)$ and $s_{p^{\text{post}}}^*(V^{\text{IP}}) = s_{\rho}^*(\bar{V}(\rho))$.

phisticated” researcher. “Sophisticated” means that the researcher correctly predicts her future behavior: she anticipates that, once an idea arrives, implementation will be evaluated under p^{post} rather than p^{pre} . She is not assumed to debias her current beliefs fully; sophistication concerns forecasting future behavior, not eliminating the underlying projection distortion.

If the researcher anticipates that upon idea arrival her belief for the implementation decision will shift to p^{post} , the continuation value $\bar{V}^{\text{so}}$ is characterized by the fixed point of the following Bellman equation.

$$\bar{V}^{\text{so}} = \delta \int_{s \geq s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})} \left[V_{p^{\text{pre}}}^*(s) - c + \delta(\bar{V}^{\text{so}} - k) \right] \bar{g}_{p^{\text{pre}}}(s) ds + \delta(\bar{V}^{\text{so}} - k) \bar{G}_{p^{\text{pre}}}(s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})).$$

Let $\bar{V}^{\text{naïve}} := \bar{V}(p^{\text{pre}})$ denote the naïve IP researcher’s continuation value. We can show $\bar{V}^{\text{naïve}} > \bar{V}^{\text{so}}$, and hence $s_{p^{\text{post}}}^*(\bar{V}^{\text{naïve}}) > s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})$. Thus, naïveté leads the IP researcher to be more selective and less likely to quit. The proof is provided in Appendix B.

3.3 Comparison of Consistent-Misbelief and Information-Projection Researchers

From the behavior of CM and IP researchers characterized above, we obtain the following result:

Proposition 1. *An IP researcher with $(p^{\text{pre}}, p^{\text{post}})$ is more likely to draw a new idea — equivalently, less likely to implement the current idea — and less likely to quit than any CM researcher with $\rho \in [p^{\text{pre}}, p^{\text{post}}]$.*

The intuition of Proposition 1 is as follows. Relative to a CM type with $\rho \in [p^{\text{pre}}, p^{\text{post}}]$, an IP researcher evaluates the *current* idea using p^{post} pessimistically, since an idea in hand is perceived as more likely to have been had before. In contrast, the IP researcher evaluates *future* ideas using p^{pre} optimistically, since ideas not yet in hand are perceived as more likely to be unique. This raises the implementation cutoff and makes drawing a new idea relatively more attractive, while also making quitting less attractive due to the higher perceived continuation value from future ideas.

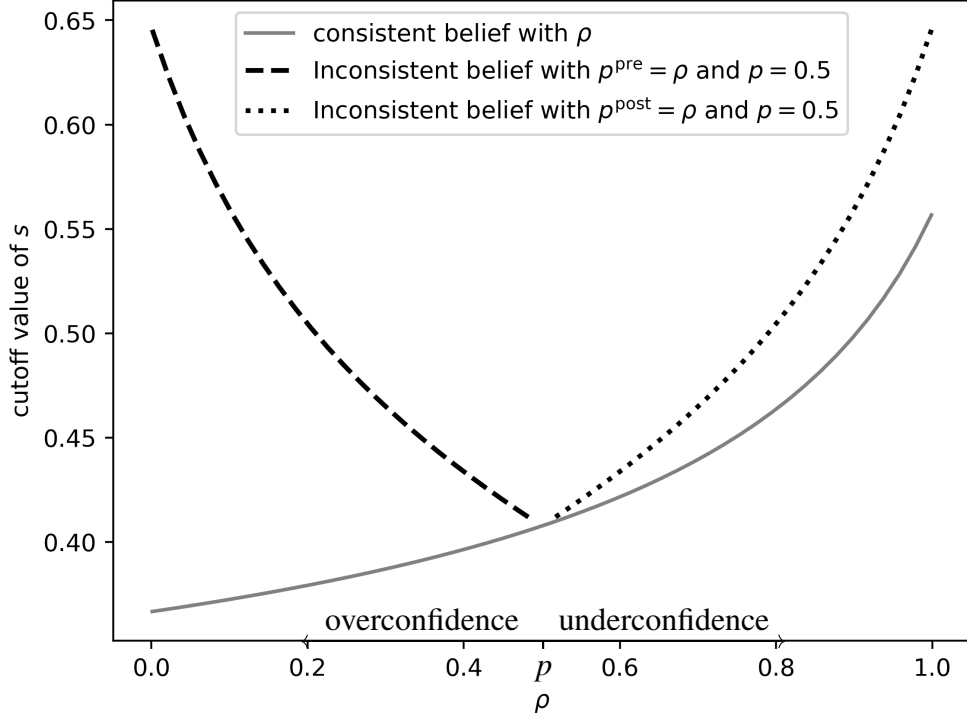


Figure 2: Numerical example of Proposition 1

The dashed curve is downward-sloping because, on that branch, the horizontal axis indexes $p^{\text{pre}} = \rho$; as ρ increases toward p , the projection weakens (α decreases) and the IP cutoff converges toward the CM cutoff. The dashed curve is the reflection of the dotted curve about $\rho = p$.

Figure 2 plots the implementation cutoffs as functions of ρ . The solid line shows the cutoff for the CM researcher. The dashed line shows the cutoff for the IP researcher when projection is indexed by $p^{\text{pre}} = \rho$, which corresponds to $\alpha = 1 - \rho/p$. The dotted line shows the cutoff for the IP researcher when we index projection by setting $p^{\text{post}} = \rho$, which corresponds to $\alpha = (\rho - p)/(1 - p)$. Under the first indexing, lower ρ corresponds to a stronger projection (larger α); under the second indexing, higher ρ corresponds to a stronger projection. Note that ρ serves only as a convenient index on the horizontal axis; the three curves need not be interpreted as arising from the same belief simultaneously.

3.4 Accomplishment and Selection

A further implication of Proposition 1 concerns which types of researchers are more likely to become accomplished: for example, by winning a prize, obtaining a job, or receiving a promotion. Our results indicate that IP researchers are disproportionately represented among those who succeed. We now investigate this mechanism formally.

We define a researcher as *accomplished* if the realized outcome of an implemented idea exceeds a threshold $X \in \mathbb{R}$, i.e., $v \geq X$. We compare two classes of types: (i) a consistent-misbelief (CM) type with constant belief ρ , and (ii) a (naïve) information-projection (IP) type characterized by $(p^{\text{pre}}, p^{\text{post}})$; equivalently, by a bias parameter $\alpha \in (0, 1)$ via $p^{\text{pre}} = (1 - \alpha)p$, $p^{\text{post}} = (1 - \alpha)p + \alpha$. Throughout this subsection, we evaluate all probabilities under the *objective* probabilities (f, g, p) : The signal observed by the researcher follows the “surviving” distribution function $\bar{G}(s) := \bar{G}_p(s)$ with its probability density function $\bar{g}(s) := \bar{g}_p(s)$. This distribution is conditional on not having been implemented by a *precursor* as mentioned in Section 3.1.

Let $s_\rho^* := s_\rho^*(\bar{V}(\rho))$ denote the cutoff for the CM type, which follows from (1) and (2). For the IP type, the current decision depends on p^{post} while the continuation value is $\bar{V}(p^{\text{pre}})$. The corresponding cutoff is $s_{\text{IP}}^* := s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}}))$. By Proposition 1, $s_{\text{IP}}^* > s_\rho^*$ for any $\rho \in [p^{\text{pre}}, p^{\text{post}}]$.

Given a cutoff s^* , define

$$q(s^*) := 1 - \bar{G}(s^*), \quad r(s^*; X) := \Pr(v \geq X \mid s \geq s^*, \text{ survival}) = \frac{\int_{s^*} \left[\int_X \varphi_p(v, s) dv \right] ds}{1 - \bar{G}(s^*)}.$$

$q(s^*)$ is the per-draw probability of implementation (conditional on survival), and $r(s^*; X)$ is the success probability conditional on implementing. Under the MLRP, $q(\cdot)$ is decreasing and $r(\cdot; X)$ is increasing in s^* .

To study selection in a nondegenerate way, this subsection augments the stationary cutoff policy with an exogenous per-period career-survival probability δ . This is not a replacement for the earlier endogenous continuation condition in the Bellman problem. Rather, it is an additional device that allows us to define an eventual-accomplishment probability and compare the composition of

researchers who pass a high threshold. Without such an exogenous hazard, eventual success can become degenerate in an infinite-horizon stationary environment whenever continuation remains optimal.¹³

The probability of eventually becoming accomplished under a stationary policy with cutoff s^* , denoted $W(s^*; X)$, satisfies

$$\begin{aligned} W(s^*; X) &= \delta q(s^*) \left[r(s^*; X) + (1 - r(s^*; X)) \delta W(s^*; X) \right] + \delta (1 - q(s^*)) W(s^*; X) \\ &= \frac{\delta q(s^*) r(s^*; X)}{1 - \delta(1 - q(s^*)) - \delta^2 q(s^*)(1 - r(s^*; X))}. \end{aligned}$$

The recursion is straightforward: with probability δq there is an implementation next period and success occurs with probability r ; otherwise, with probability $\delta q(1 - r)$ the process continues two periods ahead, and with probability $\delta(1 - q)$ it continues one period ahead.

We can now state the selection result.

Proposition 2 (Selection into accomplishment). *Assume the standard tail regularity: for any $s > s'$, $\lim_{v \rightarrow \infty} g(s | v) / g(s' | v) = \infty$.¹⁴ Then there exists $\bar{X}$ such that for all $X \geq \bar{X}$,*

$$W(s_{\text{IP}}^*; X) \geq \sup_{\rho \in [p^{\text{pre}}, p^{\text{post}}]} W(s_{\rho}^*; X),$$

with strict inequality unless the distributions are degenerate.

Proposition 2 indicates that for sufficiently high accomplishment thresholds X , the share of IP researchers among those who eventually become accomplished ($v \geq X$) is (weakly) higher than that of any CM type, relative to the ex-ante population. We discuss two issues concerning Proposition 2 in the following.

Heterogeneous IP researchers In the baseline comparison above, Proposition 2 implies that the accomplished pool contains a higher fraction of IP types when the performance bar X is high.

¹³Indeed, a CM researcher quits if and only if $\bar{V}(\rho) < k$, and an IP researcher quits if and only if $\bar{V}(p^{\text{pre}}) < k$. This is independent of the calendar time and the signal value.

¹⁴For example, normal distributions satisfy the condition.

Allowing heterogeneity in IP strength α across researchers yields a within-IP selection: for large X , $W(s_{\text{IP}}^*(\alpha); X)$ is (strictly) increasing in α , so the distribution of α among accomplished researchers first-order stochastically dominates that in the ex-ante population.

Learning and selection A natural question is whether researchers learn about p over time. Our baseline abstracts from Bayesian learning to isolate the mechanism whereby ideas are evaluated with state-dependent beliefs. Importantly, even if some researchers learned and converged to a stationary unbiased cutoff s_p^* that is *weakly below* the IP cutoff $s_{\text{IP}}^* = s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}}))$, our selection result implies that such learners would be *under-represented* among accomplished researchers when the bar X is high.

Formally, by Proposition 1 we have $s_{\text{IP}}^* > s_\rho^*$ for any CM belief $\rho \in [p^{\text{pre}}, p^{\text{post}}]$; in particular $s_{\text{IP}}^* > s_p^*$. Proposition 2 shows that, for sufficiently large X , the eventually being accomplished probability $W(s^*; X)$ is strictly increasing in s^* by the MLRP and a standard tail regularity condition. Hence for any type with $s_L^* \leq s_p^* < s_{\text{IP}}^*$ we have $W(s_L^*; X) < W(s_{\text{IP}}^*; X)$. Therefore, conditional on becoming accomplished ($v \geq X$), the cross-sectional composition tilts toward IP types. Intuitively, the ex-post pessimism of IP researchers raises the implementation cutoff; for high thresholds X , “fewer but higher- s ” implementations yield a higher success probability, which selection amplifies.

4 Conclusion

This study considers a researcher who faces a trade-off between implementing a current research idea and drawing a new one over an infinite horizon. The researcher is uncertain about the value of the current idea and believes that other researchers may already have drawn the same one. By developing a search model, we show that researchers with information projection bias are more selective than those with overconfidence or underconfidence: the former are more likely to discard the current idea and draw a new one.

The key mechanism distinguishing information projection bias from other biases is uncertainty about an idea’s uniqueness, which arises from the file drawer problem. This problem undermines

researchers' confidence, making them more selective and less motivated to pursue research. Our study shows that projection bias influences the file drawer problem in two distinct ways. First, from an ex-ante perspective, information projection bias increases researchers' willingness to conduct research by making them more optimistic. Second, from an ex-post perspective, it makes researchers more pessimistic, leading them to be more selective in the ideas they choose to pursue. The tension between these opposing effects of information projection bias has a significant impact on researchers' decision-making and overall performance.

Declaration

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Appendix A. Proofs

Proof of Lemma 1. First, we note that $\frac{\varphi_\rho(v, s)}{\varphi_\rho(v, s')} = \frac{g(s|v)}{g(s'|v)}$ increases in v if and only if $s > s'$ by the MLRP of G .

Similarly, we show the following properties.

Claim 1. $\frac{\varphi_\rho(v|s)}{\varphi_{\rho'}(v|s)}$ increases in v if and only if $\rho' > \rho$.

Proof of Claim 1. By definition,

$$\frac{\varphi_\rho(v, s)}{\varphi_{\rho'}(v, s)} = \frac{1 - \rho \cdot [1 - G(\bar{s} | v)]}{1 - \rho' \cdot [1 - G(\bar{s} | v)]}.$$

Then, $\frac{\varphi_\rho(v,s)}{\varphi_{\rho'}(v,s)}$ increases in v if and only if

$$-\frac{\partial G(\bar{s} | v)}{\partial v}(\rho' - \rho) > 0.$$

As the MLRP implies the first-order stochastic dominance, $\frac{\partial G(\bar{s}|v)}{\partial v} < 0$. Therefore, $\frac{\varphi_\rho(v,s)}{\varphi_{\rho'}(v,s)}$ increases in v if and only if $\rho' > \rho$. $\square$

This implies the MLRP of φ for smaller ρ and higher s . Then, the statement immediately follows from its first-order stochastic dominance property. $\square$

Proof of Lemma 3. To simplify the notation, let $w = \int [1 - \rho(1 - G(\bar{s} | v))]f(v)dv$ and $w_s = \int [1 - \rho(1 - G(\bar{s} | v))]G(s | v)f(v)dv$. By denoting $X = 1 - G(\bar{s} | v)$ and $Y = G(s | v)$, we can see that X and Y are random variables. Then, we have that

$$\begin{aligned} \frac{\partial \bar{G}_\rho(s)}{\partial \rho} &= -w^{-2} \int (1 - G(\bar{s} | v))[wG(s | v) - w_s]f(v)dv = -w^{-2}(E[XY] - E[X]E[Y]) \\ &= -w^{-2}\text{Cov}(X, Y). \end{aligned}$$

Note that $G(\cdot | v)$ is strictly decreasing in v by the MLRP while X is increasing in v , Y is decreasing in v , which implies that $\text{Cov}(X, Y) < 0$ and thus, $\frac{\partial \bar{G}_\rho(s)}{\partial \rho} > 0$. $\square$

Proof of Lemma 4. The proof consists of two parts.

Existence and Uniqueness Let $\Psi_\rho(\tilde{s})$ be a function defined by

$$\Psi_\rho(\tilde{s}) := \frac{\delta \int_{\tilde{s}} \bar{g}_\rho(s) [V_\rho^*(s) - c] ds - k [\delta^2 + \delta(1 - \delta)\bar{G}_\rho(\tilde{s})]}{1 - \delta^2 - \delta(1 - \delta)\bar{G}_\rho(\tilde{s})}.$$

Then, when $\bar{V} = \Psi_\rho(s_\rho^*(\bar{V}))$, the Bellman equation (2) holds. We show a property of Ψ_ρ .

Claim 2. $\Psi'_\rho(\tilde{s}) > 0$ if and only if $\bar{V} < \Psi_\rho(\tilde{s})$ at $\tilde{s} = s_\rho^*(\bar{V})$.

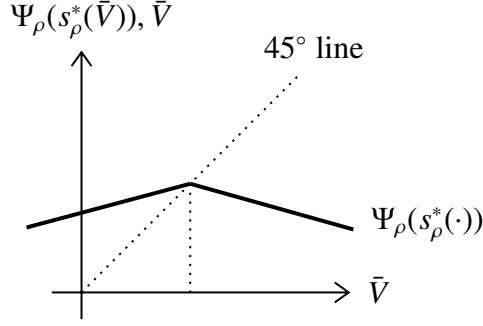


Figure 3: Graph of $\Psi_\rho(s_\rho^*(\cdot))$

Proof of Claim 2. By differentiating $\Psi_\rho(\tilde{s})$ with respect to $\tilde{s}$, $\Psi'_\rho(\tilde{s}) > 0$ if and only if

$$\delta[(V_\rho^*(\tilde{s}) - c) + k(1 - \delta)] < (1 - \delta)\delta \frac{\left[\delta \int_{\tilde{s}} \bar{g}_\rho(s) [V_\rho^*(s) - c] ds - k [\delta^2 + \delta(1 - \delta)\bar{G}_\rho(\tilde{s})] \right]}{(1 - \delta^2 - \delta(1 - \delta)\bar{G}_\rho(\tilde{s}))}. \quad (3)$$

From (1),

$$V_\rho^*(s_\rho^*(\bar{V})) - c + (1 - \delta)k = (1 - \delta)\bar{V}.$$

The inequality (3) is equivalent to $\bar{V} < \Psi_\rho(s_\rho^*(\bar{V}))$. □

As $s_\rho^*(\bar{V})$ increases in $\bar{V}$, the shape of $\Psi_\rho(s_\rho^*(\bar{V}))$ is drawn as shown in Figure 3. This shows the uniqueness of the solution to $\bar{V} = \Psi_\rho(s_\rho^*(\bar{V}))$. Let $\bar{V}(\rho)$ be the unique solution.

Comparative statics with respect to ρ Consider the case where $\bar{V}(\rho) > k$. We prepare the following claim.

Claim 3. Let $w: \mathbb{R} \rightarrow \mathbb{R}_+$ be a continuously differentiable function with $w' > 0$ and $\sup w < \infty$.

Define

$$\tilde{\Psi}_\rho(\tilde{s} | w) := \frac{\delta \int_{\tilde{s}} w(s) \bar{g}_\rho(s) ds - k [\delta^2 + \delta(1 - \delta)\bar{G}_\rho(\tilde{s})]}{1 - \delta^2 - \delta(1 - \delta)\bar{G}_\rho(\tilde{s})}.$$

Then, $\frac{\partial \tilde{\Psi}_\rho(\tilde{s})}{\partial \rho} < 0$.

Proof of Claim 3. Note that $\frac{\partial \tilde{\Psi}_\rho(\tilde{s}|w)}{\partial \rho} < 0$ if and only if

$$\begin{aligned} & \delta \int_{\tilde{s}} w(s) \frac{\partial \bar{g}_\rho(s)}{\partial \rho} ds \left[1 - \delta^2 - \delta(1 - \delta) \bar{G}_\rho(\tilde{s}) \right] + \left[\delta \int_{\tilde{s}} w(s) \bar{g}_\rho(s) ds - k \right] \delta(1 - \delta) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} < 0 \\ \Rightarrow & \delta \int_{\tilde{s}} w(s) \left[\frac{\partial \bar{g}_\rho(s)}{\partial \rho} \left[1 - \delta^2 - \delta(1 - \delta) \bar{G}_\rho(\tilde{s}) \right] + \delta(1 - \delta) \bar{g}_\rho(s) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} \right] ds - k \delta(1 - \delta) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} < 0. \end{aligned}$$

Using the integral by parts,

$$\begin{aligned} Z &:= \int_{\tilde{s}} w(s) \left[\frac{\partial \bar{g}_\rho(s)}{\partial \rho} \left[1 - \delta^2 - \delta(1 - \delta) \bar{G}_\rho(\tilde{s}) \right] + \delta(1 - \delta) \bar{g}_\rho(s) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} \right] ds \\ &= \left[w(s) \left[\frac{\partial \bar{G}_\rho(s)}{\partial \rho} \left[1 - \delta^2 - \delta(1 - \delta) \bar{G}_\rho(\tilde{s}) \right] + \delta(1 - \delta) \bar{G}_\rho(s) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} \right] \right]_{\tilde{s}}^{\infty} \\ &\quad - \int_{\tilde{s}} w'(s) \left[\frac{\partial \bar{G}_\rho(s)}{\partial \rho} \left[1 - \delta^2 - \delta(1 - \delta) \bar{G}_\rho(\tilde{s}) \right] + \delta(1 - \delta) \bar{G}_\rho(s) \frac{\partial \bar{G}_\rho(\tilde{s})}{\partial \rho} \right] ds. \end{aligned}$$

As $\lim_{s \rightarrow \infty} \bar{G}_\rho(s) = 1$ for any ρ , $\lim_{s \rightarrow \infty} \frac{\partial \bar{G}_\rho(s)}{\partial \rho} = 0$. As w is bounded above and $\frac{\partial \bar{G}_\rho(s)}{\partial \rho} \geq 0$ by Lemma 3, $Z \leq 0$. Therefore, $\frac{\partial \tilde{\Psi}_\rho(\tilde{s}|w)}{\partial \rho} \leq 0$. $\square$

Note that, as u is bounded above and $V_\rho^*(\cdot)$ is an expectation of u , $V_\rho^*(\cdot)$ is also bounded above. Also, $V_\rho^*(\cdot)$ is increasing in s by Lemma 1. Due to the cut-off point defined by (1), we also have $V_\rho^*(s) > c$ for each $s \geq s_\rho^*(\bar{V})$. As the cutoff point is continuous in ρ , $V_\rho^*(s) > c$ for each $s \geq s_{\rho'}^*(\bar{V})$ when ρ' is sufficiently close to ρ . Now take ρ and ρ' with $\rho > \rho'$, when ρ' is sufficiently close to ρ , for any $\tilde{s} \geq \min\{s_{\rho'}^*(\bar{V}), s_\rho^*(\bar{V})\}$, Claim 3 and Lemma 1 (i) imply that

$$\Psi_\rho(\tilde{s}) = \tilde{\Psi}_\rho(\tilde{s} | V_\rho^*(\tilde{s})) \leq \tilde{\Psi}_{\rho'}(\tilde{s} | V_\rho^*(\tilde{s})) \leq \tilde{\Psi}_{\rho'}(\tilde{s} | V_{\rho'}^*(\tilde{s})).$$

Then, we have that $\frac{\partial \tilde{\Psi}_\rho(\tilde{s})}{\partial \rho} < 0$.

Note that by Claim 2 and with $\bar{V}(\rho)$ as the fixed point of Ψ_ρ , $\Psi'_\rho(s_\rho^*(\bar{V}(\rho))) = 0$. Therefore, by the total differentiation of $\Psi_\rho(s_\rho^*(\bar{V}(\rho))) = \bar{V}(\rho)$ with respect to ρ , we have $\bar{V}'(\rho) < 0$. $\square$

Proof of Proposition 1. We compare an IP researcher with a CM researcher. Recall that the IP researcher's cutoff on the decision of implementation is $s_{p^{\text{pre}}}^*(\bar{V}(p^{\text{pre}}))$ while the CM researcher's

cutoff is $s_\rho^*(\bar{V}(\rho))$. From Lemma 4, $\bar{V}(p^{\text{pre}}) \geq \bar{V}(\rho)$ for any $\rho \in [p^{\text{pre}}, p^{\text{post}}]$ if $\bar{V}(\rho) > k$. As $s_\rho^*(\bar{V})$ increases in ρ and $\bar{V}$ by Lemma 2, $s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}})) > s_\rho^*(\bar{V}(\rho))$ for any $\rho \in [p^{\text{pre}}, p^{\text{post}}]$.

Next, consider the decision on whether to quit. The IP researcher quits the research if and only if $\bar{V}(p^{\text{pre}}) < k$ while the CM researcher does if $\bar{V}(\rho) < k$. As $\bar{V}(p^{\text{pre}}) \geq \bar{V}(\rho)$ for any $\rho \in [p^{\text{pre}}, p^{\text{post}}]$ if $\bar{V}(\rho) > k$, the IP researcher is less likely to quit. $\square$

Proof of Proposition 2. By Proposition 1, $s_{\text{IP}}^* > s_\rho^*$ for any $\rho \in [p^{\text{pre}}, p^{\text{post}}]$.

$\frac{\partial W(s^*; X)}{\partial s^*} > 0$ if and only if

$$\begin{aligned} & - \left[\int_X \varphi_p(v, s^*) dv \right] \left[1 - \delta(1 - q(s^*)) + \delta^2 \left[q(s^*) - \int_{s^*} \int_X \varphi_p(v, s) dv ds \right] \right] \\ & > \left[\delta(1 + \delta)q'(s^*) + \delta^2 \int_X \varphi_p(v, s^*) dv \right] \left[\int_{s^*} \int_X \varphi_p(v, s) dv ds \right] \\ & \Rightarrow \frac{(1 - \delta) + \delta(1 + \delta)q(s^*)}{\delta(1 + \delta)\bar{g}(s^*)} < \int_{s^*} \frac{\int_X \varphi_p(v, s) dv}{\int_X \varphi_p(v, s^*) dv} ds. \end{aligned} \quad (4)$$

Note that $\frac{\int_X \varphi_p(v, s) dv}{\int_X \varphi_p(v, s^*) dv}$ is a convex combination of $\frac{\varphi_p(v, s)}{\varphi_p(v, s^*)} = \frac{g(s|v)}{g(s^*|v)}$. For each $s > s^*$, when $X \rightarrow \infty$, $\frac{g(s|v)}{g(s^*|v)} \rightarrow \infty$ for each $v \geq X$ by the tail regularity. Then, $\lim_{X \rightarrow \infty} \frac{\int_X \varphi_p(v, s) dv}{\int_X \varphi_p(v, s^*) dv} = \infty$ for each $s > s^*$ and thus $\lim_{X \rightarrow \infty} \int_{s^*} \frac{\int_X \varphi_p(v, s) dv}{\int_X \varphi_p(v, s^*) dv} ds = \infty$. Therefore, inequality (4) is satisfied for large X , which implies that $\frac{\partial W(s^*; X)}{\partial s^*} > 0$. $\square$

Appendix B. Sophisticated researcher

In the main model, the IP researcher is assumed to be naïve: she does not expect her belief to shift over time. In this section, we consider the case in which an IP researcher rationally anticipates this shift. More precisely, the *sophisticated* IP researcher has the same information-projection bias as the naïve type. However, even before drawing an idea, she realizes that her future self will decide based on the updated belief after a future idea is drawn.¹⁵ To distinguish the two types, we refer to the IP researcher in the main model as the *naïve* IP researcher.

¹⁵We can also consider an agent who anticipates and fully internalizes her belief change ex ante; this collapses to a special case of consistent misbelief with $\rho = p^{\text{post}}$.

The sophisticated IP researcher anticipates that her belief will update to p^{post} upon idea arrival, and the implementation decision is based on p^{post} . By contrast, because her continuation is evaluated under the ex-ante belief p^{pre} , the continuation value is characterized by the fixed point of

$$\bar{V}^{\text{so}} = \delta \int_{s \geq s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})} \left[V_{p^{\text{pre}}}^*(s) - c + \delta (\bar{V}^{\text{so}} - k) \right] \bar{g}_{p^{\text{pre}}}(s) ds + \bar{G}_{p^{\text{pre}}}(s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})) \delta (\bar{V}^{\text{so}} - k). \quad (5)$$

Note that $s_{p^{\text{post}}}^*$ satisfies $V_{\rho}^*(s_{\rho}^*(\bar{V}^{\text{so}})) = (1 - \delta)(\bar{V}^{\text{so}} - k) + c$, which is the solution to (1) at $\rho = p^{\text{post}}$. This follows from the assumption that a sophisticated researcher anticipates that her future implementation decision will be made by her future self, who holds belief p^{post} . As in the naïve IP researcher's case, the sophisticated researcher evaluates the current idea under the ex-post belief p^{post} ; therefore, her current-period implementation decision is determined by the cutoff $s_{p^{\text{post}}}^*(\bar{V}^{\text{so}})$. We then obtain the following result.

Proposition 3. *The naïve IP researcher is more likely to draw a new idea and less likely to quit than the sophisticated IP researcher.*

Proof. Compare the naïve IP researcher with the sophisticated IP researcher. Let $\bar{V}^{\text{so}}(p^{\text{pre}})$ denote the solution to (5), with the cutoff $s_{p^{\text{post}}}^*(\cdot)$ determined by (1) evaluated at $\rho = p^{\text{post}}$. We have $\bar{V}^{\text{so}}(p^{\text{pre}}) < \bar{V}(p^{\text{pre}})$ because $s^* = s_{p^{\text{post}}}^*(\bar{V})$ is not optimal for the researcher who evaluates continuation under p^{pre} ; hence the sophisticated researcher is more likely to quit. Moreover, $\bar{V}(p^{\text{pre}}) > \bar{V}^{\text{so}}(p^{\text{pre}})$ implies $s_{p^{\text{post}}}^*(\bar{V}(p^{\text{pre}})) > s_{p^{\text{post}}}^*(\bar{V}^{\text{so}}(p^{\text{pre}}))$, so the sophisticated researcher's cutoff is lower, which implies that she is less selective. $\square$